Revision 1:
System Requirements and Hazard Analysis

Group 8 - C.A.V.E: Cave Assessment and Visualization Equipment

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Table 0: Revision History

Date	**Developer(s)**	**Change**
November 16, 2025	All Members	Initial Draft
March 18, 2026	All Members	Update requirements based on V&V. Remove references to LiDAR. Improve system diagram.
March 18, 2026	Tabish Faisal	Re-worded Hardware requirements to be singular instead of plural (1 ToF instead of 2 ToFs). Updated Software Analysis to better match Hardware Analysis. Included System Requirements that relate to Software and Hardware Hazard Analyses.

Contents

1 Introduction

1.1 Purpose

The purpose of this document is to outline the system requirements of the C.A.V.E. system, highlighting expected behaviours such as the inputs collected, what type of processing is performed on the data, and outputs generated. It will define all the necessary variables and constants used to achieve this, as well as any notation or technical terms that are needed. There will also be a fault tree analysis (FTA) breakdown of the possible root causes of failure that will inform the significant safety requirements of the system.

1.2 Scope

The scope of the system includes the development of a portable or wearable 3D data-acquisition device capable of reconstructing a scanned environment with integrated sensors and processing algorithms. The system records sequential point clouds from the Time-of-Flight sensor, fuses IMU orientation data and processes the collected data to generate an accurate and reliable 3D scan of the environment. This scope covers the areas of data capture, pre-processing, point cloud registration, as well as hardware and software integration.

2 Project Definitions & Variables

2.1 Glossary

The following table consists of definitions of certain terms and notations that are referenced in the project.

Term	**Definition**
IMU	Inertial Measurement Unit $\rightarrow$ Provides acceleration and rotational data
Point Cloud	A collection of coordinate points in 3D space
Portable Device	The portion of the design that is carried with the user/operator for mapping an enclosed space
ToF	Time-of-Flight depth sensor $\rightarrow$ Uses pulses of infrared light and a sensor that measures the time at which the light is received back to create a point cloud

Table 1: Definition of Key Terms and Notation

2.2 Monitored and Controlled Variables

Monitored variables are information from the environment being input to the system, and controlled variables are the output products of the system's internal processes. The following tables list and describe these variables in the C.A.V.E. system.

Variable	**Description**	**Unit**	**Purpose**

M_tof_data	A single depth image frame captured from the environment	Array of depths and positions in metres	Capturing the contours and shape of the environment
M_imu_gyro	Device rotation rate about x,y,z axes, measured by IMU	Degrees/second tuple $(\deg/s_x, \deg/s_y, \deg/s_z)$	Provide reference for position change between M_tof_data captures
M_imu_accel	Device acceleration in x, y, z axes, measured by IMU	Metres/second^2 tuple $(m/s_x^2, m/s_y^2, m/s_z^2)$	Additional frame translation reference
M_imu_magnet	Change in magnetic field	Tesla tuple (T_x, T_y, T_z)	Additional frame rotation reference
M_start_stop	User toggle enabling or disabling the system	Boolean, true == on	Allow user control of starting and stopping a mapping session
M_SOC	Remaining battery percentage	Percentage of total charge (%)	Estimate remaining time available for mapping
M_storage_left	Remaining storage space available for data capture	Percentage of allocated storage (%)	Estimate remaining storage available for mapping
C_record_status	User facing indicator to show the status of M_start_stop	Boolean	Inform user of the status of M_start_stop
C_record_remain	User facing indicator to show how long recording can continue for, given M_SOC and M_storage	Integer in the form MM:SS	Inform user how much longer they can record
C_point_cloud	Final point cloud or 3D representation of the merging of all captured data	Point cloud; array of X,Y,Z tuples	Main product of the device, 3-dimensional representation of the mapped area (cave)

Table 2: Variables and its Description

2.3 Constants

This section describes the values that will not change during the operation of the device and should be set to a specific value during design.

Variable	Description	Unit	Purpose

Const_transform_i	Transformation matrix of $sensor_i$ from the reference frame (likely chosen as the IMU or centre of portable device)	4x4 matrix representing translation and rotation in 3D	Define spatial relationship between different sensor's captures
Const_poll_rate_i	Rate at which $sensor_i$ makes information available to the system	Hertz	Inform rate of data ingest
Const_capt_size_i	Size of a single capture of information by $sensor_i$	Gigabytes (GB)	Inform Const_datarate
Const_capt_rate	Reciprocal of the interval between a total system capture of all sensors	Hertz (s^{-1})	Inform Const_datarate
Const_storage	Storage capacity of the portable device	Gigabytes (GB)	Inform M_storage_left
Cosnt_datarate	Total ingested data from all sensors, the sum of Const_capt_rate * Const_capt_size_i	Gigabytes/second (GB/s)	Inform C_record_remain
Cosnt_power_rate	Average power consumption rate of the portable device	Watts (W)	Informs C_record_remain
Cosnt_power_cap	Battery capacity of the portable device	Watt-hours (Wh)	Informs C_record_remain

Table 3: Constants and its Description

3 System Diagram

In Figure 1, the system is represented as a black-box, where the monitored variables serve as the inputs to the system and the controlled variables are the resulting outputs.

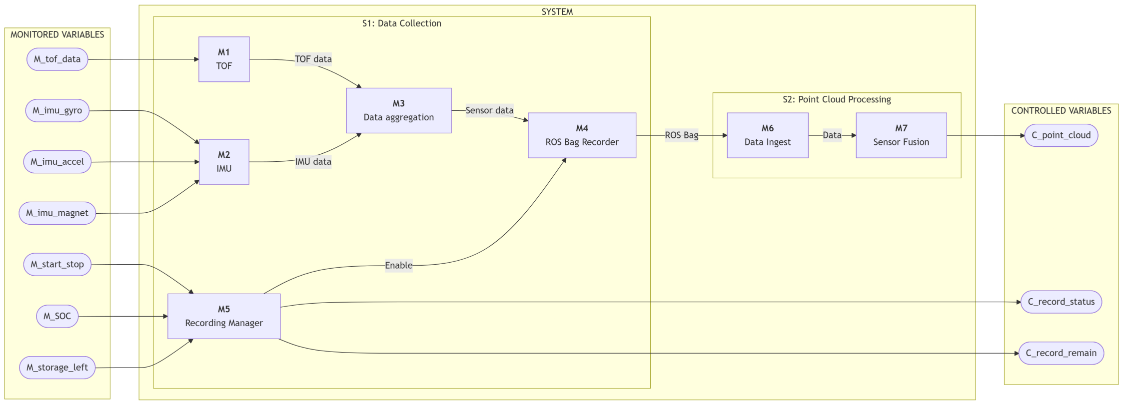

Figure 1: Representation of System Boundaries

4 Functional Decomposition

The folllowing figure shows the functional decomposition of the C.A.V.E. system.

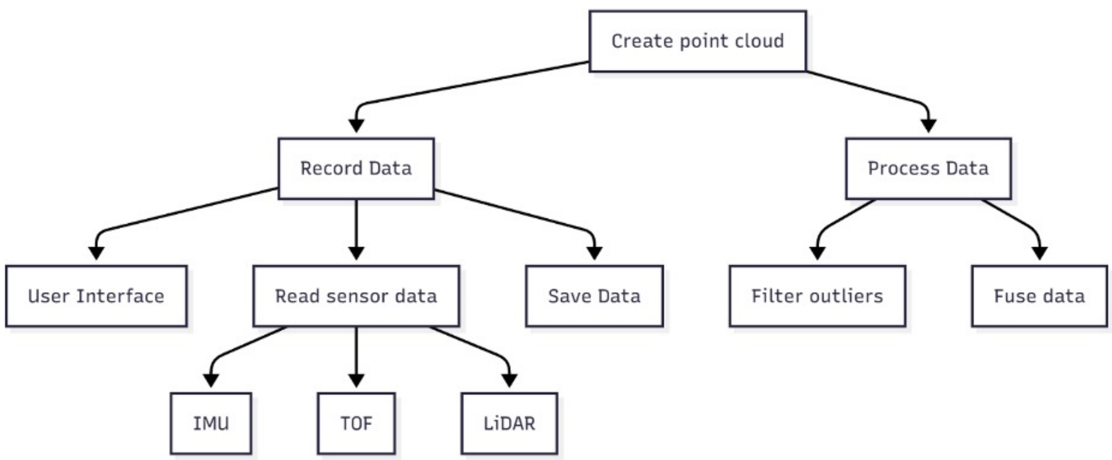

Figure 2: Functional Decomposition of the System

5 System Requirements

Requirement Type	Likelihood of Change
Behavioural	
RB1: The device will produce a point cloud C_point_cloud representing the physical surfaces mapped by the sensor (M_tof_data) while recording is enabled (M_start_stop == true).	Low - this is a core functionality of the device.

RB2: *The confidence in the accuracy of C_point_cloud shall be represented by C_confidence*	This requirement has been removed. See the Verification and Validation document for more details.
RB3: C_record_status toggles between indicating recording and standby when M_start_stop is toggled by the user. If the system encounters an error it cannot recover from, it must display an error status on C_record_status.	Low - this is a basic error communication.
RB4: C_record_time will display remaining available recording time by taking the minimum of M_SOC/const_power_rate and M_storage_left/const_datarate.	Low - Important to achieve for ease of use by the operator.
RB5: The quality of the point cloud produced by the system must be independent of all external light.	Possible - This requirement has already been modified once, and is unlikely to change further.
Timing	
RT1: Const_capt_rate shall be frequent enough to produce complete maps of an environment at a walking pace (3-5km/h)	Low - Core function needed for ease of use.
RT2: Const_capt_rate shall be low enough such that the system can complete a measurement of all sensor data before the next interval.	Low - Need to ensure consistent data.
Hardware/Software Environment	
RE1: The portable device shall not require an active internet connection to achieve the behavioural requirements RBX.	Low - Core function, must potentially operate underground.
RE2: The hardware should minimize encumbrance of the user to ensure exploration is not inhibited.	Low - Important for ease of use.
RE3: The IMU sensors data should be accurate within the accepted range.	Low - Direct requirement to mitigate Hazards S2R5-S2R8
RE4: The ToF sensor data should be accurate within the accepted range.	Low - Direct requirement to mitigate Hazards S2R1-S2R4
RE5: The sensor fusion should be accurate within the accepted range.	Low - Direct requirement to mitigate Hazard S2R11
RE6: Battery should be operating in nominal condition, without excess power draw and properly tracked and regulated capacity.	Low - Direct requirement to mitigate Hazards H2R1, H2R2, S2R9 and S2R10
RE7: Side buttons for powering on should work as expected.	Low - Direct requirement to mitigate hazards S2R14-S2R16

RE8: All hardware components shall be properly connected together.	Low - Direct requirement to mitigate hazards H2R5-H2R10, H2R13, H2R14, H2R15, H2R18, H2R21, and H2R26

Table 4: Requirement and its Likeliness to Change

6 Hazard Analysis

In this section, the safety requirements are determined from a breakdown of the possible hazards that may occur in the system with respect to both its software and hardware components. These hazards are analyzed through an FTA as shown in Figures 3 and 4.

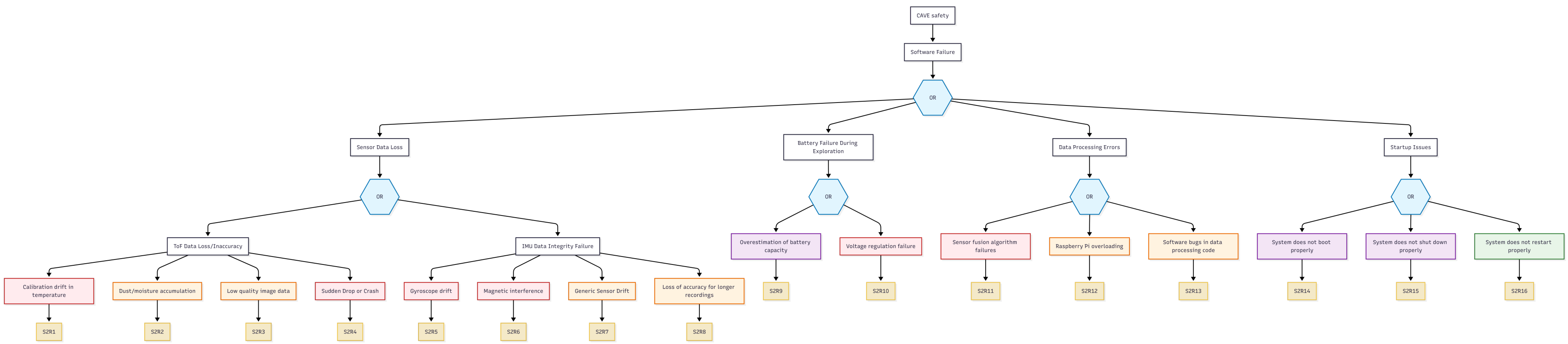

Figure 3: FTA of the software components in the system

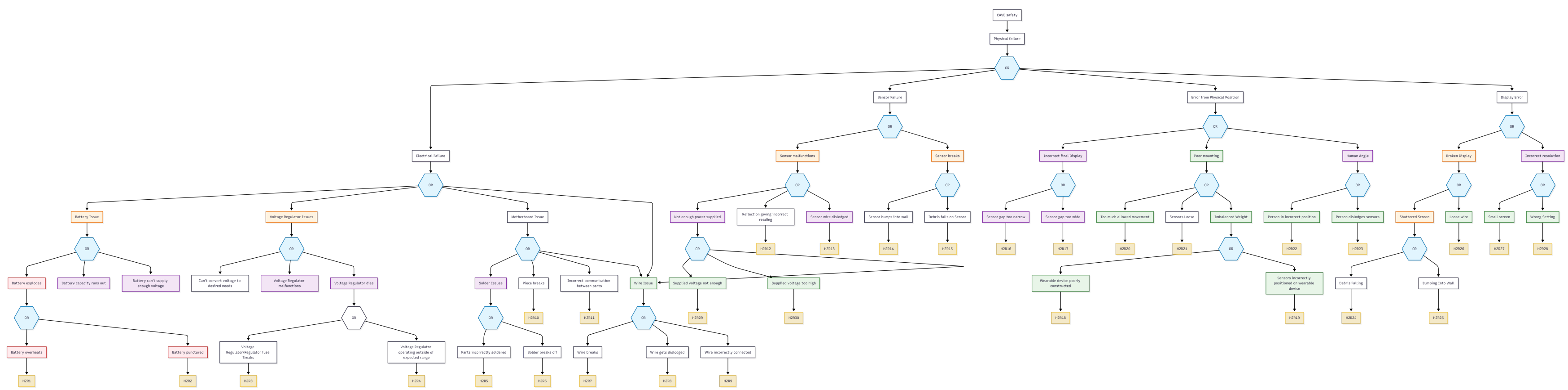

Figure 4: FTA of the hardware components in the system

6.1 Error Handling/Mitigation

The following table lists the error handling methods or mitigation when any hazards occur in the system along with its respective severity.

Hazard ID	Description	Severity	Response Strategy	Mitigation Plan
S2R1	Calibration drift in temp	Severe	Mitigate	Ensure temperatures remain in nominal range
S2R2	Dust & moisture accumulation	High	Mitigate	Proper housing for IMU to limit dust and moisture penetration
S2R3	Low quality image data	High	Mitigate	Limited from the quality of the ToF sensor itself. Ensure it doesn't hinder the overall product too much
S2R4	Sudden drops or crashes	Severe	Avoid	Ensure we don't run too long sessions with risk of losing all acquired data
S2R5	Gyroscope drift	Severe	Mitigate	Account for potential drift in code or ensure drift doesn't affect accuracy to a noticeable degree
S2R6	Magnetic interference	Severe	Mitigate	Ensure no strong magnets are around the IMU
S2R7	Generic sensor drift	High	Mitigate	Re-calibrate the IMU often to reduce any general drift
S2R8	Loss of accuracy for long recordings	High	Avoid	Run shorter sessions to keep IMU as accurate as possible
S2R9	Overestimation of battery capacity	Medium	Avoid	Take battery calculations directly from the source and operate the system as if its 5% less (if it reads 95% , assume 90%)
S2R10	Voltage regulation failure	Severe	Mitigate	Ensure robust software checks such that this doesn't occur
S2R11	Sensor fusion algorithm failure	Severe	Mitigate	Proper testing and verification of the fusion algorithms before deployment
S2R12	Raspberry Pi overloading	High	Avoid	Make sure what we are running is well within the limitations of the Pi

S2R13	Software bugs in data processing code	High	Mitigate	Remove all sources of bugs to the best of our abilities such that it won't impede the safety of the device
S2R14	System does not boot properly	Medium	Avoid	Ensure foolproof solution to booting
S2R15	System does not shut down properly	Medium	Avoid	Ensure foolproof solution to shutting down
S2R16	System does not restart properly	Low	Avoid	Rigorous testing to make sure it restarts as expected
H2R1	The battery overheats	High	Mitigate	Proper cooling, proper rated battery
H2R2	Battery punctured	High	Avoid	Proper casing, kept in a place unlikely to get impacted
H2R3	Voltage regulator fuse breaks	Medium	Mitigate	Proper rating of a regulator so it doesn't fall outside of expected range
H2R4	Voltage regulator operating outside of expected range	Medium	Mitigate	Make sure no sudden power draws occur
H2R5	Parts incorrectly soldered	Medium	Avoid	Double-check soldered equipment
H2R6	Solder breaks off	Medium	Avoid	Use ample amount of solder
H2R7	Wire breaks	Medium	Avoid	Ensure wire is taut and flexible, double-check connections
H2R8	Wire gets dislodged	Medium	Avoid	Double-check wire is securely connected to different pieces, stress-test connection
H2R9	Wire improperly connected	Medium	Avoid	Double-check what the wires are connecting, make sure it follows a plan
H2R10	Motherboard component breaks	Medium	Avoid	Keep motherboard secure and in a place less prone to bumps/potentially breaking

H2R11	Motherboard parts incorrectly communicate	Medium	Avoid	Double-check motherboard connections, ensure everything is wired as expected.
H2R12	TOF sensor providing point cloud of a reflection (e.g water)	Medium	Mitigate	Ensure the other sensor can catch errors and correct them in post
H2R13	Sensor wire dislodged	Medium	Avoid	Provide the sensor enough support so the wire wouldn't accidentally break or get damaged
H2R14	Sensor bumps into wall	Medium	Avoid	Ensure sensor has proper protection so it's not directly in contact with cave walls
H2R15	Debris falls on sensor	Medium	Avoid	Ensure sensor is inset, or protected from potential vertical hazards
H2R16	Sensor gap too narrow	Medium	Avoid	Correct planning to ensure optimal sensor placements such that there is minimal overlap to produce efficient imaging
H2R17	Sensor gap too wide	Medium	Avoid	Correct planning to ensure optimal sensor placements such that there is minimal overlap to produce efficient imaging
H2R18	Wearable device poorly constructed	Low	Avoid	Proper simulation before printing, as well as rigorous testing to make sure it's constructed well
H2R19	Sensor incorrectly positioned	Low	Avoid	Mirror sensor positions so that weight is balanced
H2R20	Too much allowed movement (for sensor)	Low	Avoid	Limit degrees of freedom to 0 or 1 for each sensor
H2R21	Sensor Loose	Medium	Avoid	Proper machining such that the sensor fit snugly, without enough space to wiggle around.

H2R22	Person in incorrect position	Low	Accept	Explain to the user how to be properly angled, but accept there will be some human error in the beginning
H2R23	Person dislodges sensor	Low	Accept	Understand that there are human errors, and they may push some things incorrectly. Proper machining and fixtures will mitigate this as much as possible
H2R24	Debris Falling, shattering screen	Medium	Mitigate	Give the screen a protector so that it wouldn't be harmed to a degree of breaking
H2R25	Bumping into a wall, shattering screen	Medium	Mitigate	Give the screen proper protection so it wouldn't get damaged to such a degree
H2R26	Loose wire for the display	Low	Avoid	Proper connection between the display and other hardware, make sure there is enough wire such that it doesn't snap or pull off
H2R27	Screen too small	Low	Avoid	Research prior for assumed size of screen, such that everything would fit as expected
H2R28	Screen resolution incorrect	Low	Avoid	Make sure the screen used supports the resolution we have to display properly
H2R29	Under-supplied voltage	Low	Avoid	Shut down the system if power is insufficient
H2R30	Over-supplied voltage	Low	Avoid	Shut down the system if prolonged over-supplied voltage, to make sure parts don't break

Table 5: Hazards and its Mitigation Plan